



IN-CLASS EXERCISES



SEMANTIC ANALYSIS 4 —WRAPPING UP

Download exercises

1. Go to

<https://ligerlabs.org/compilers.html>

2. Download the file

`sem-4-exercises.zip`

3. Open up a terminal and Unzip the file

```
> unzip sem-4-exercises.zip
```

```
> cd sem-4-exercises
```

```
> ls
```

INTRODUCTION

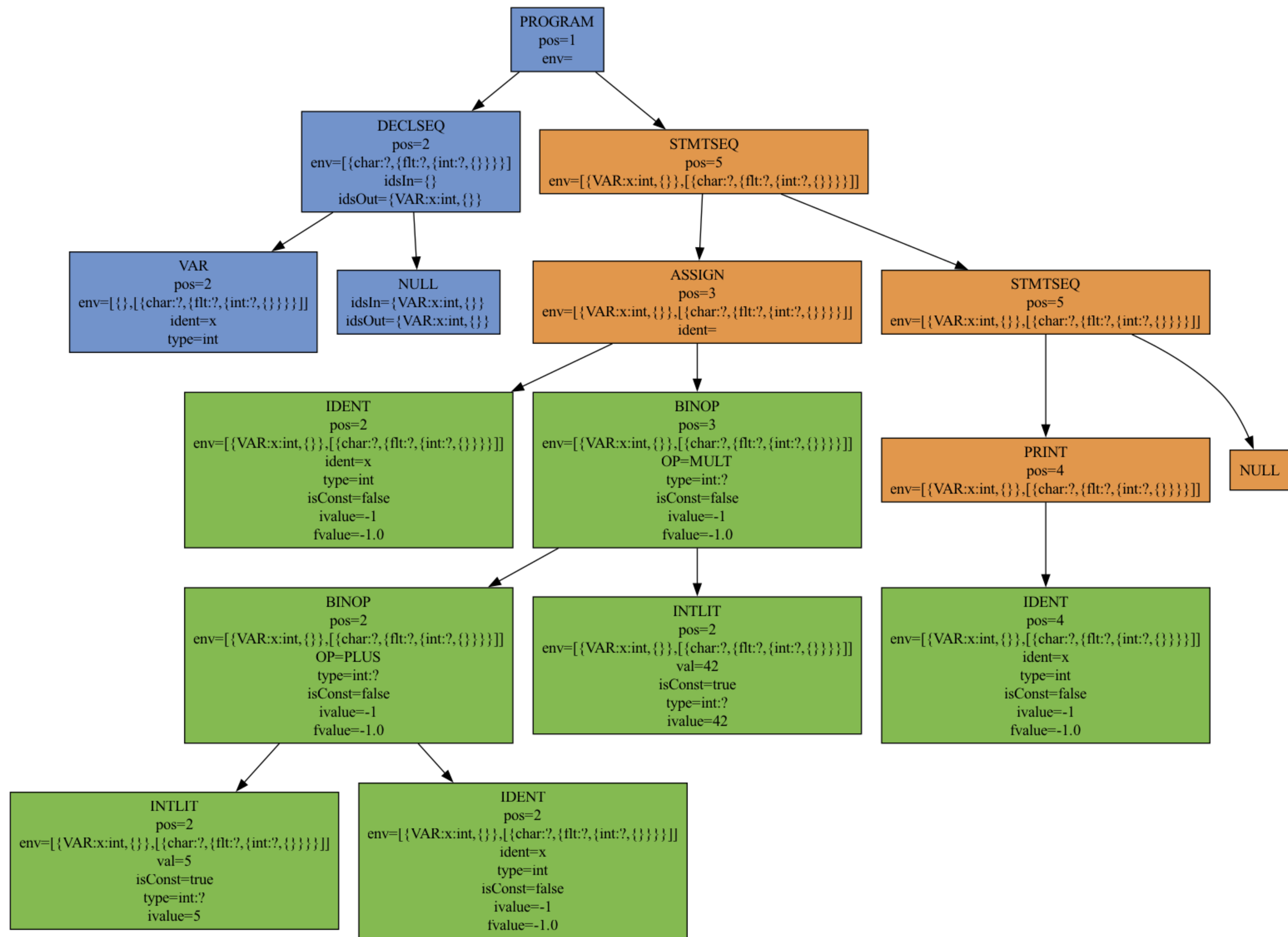
- Here's an example of a program that has a reasonably complex expression:

expr.tn

```
BEGIN
  VAR x : int;
  x = 5 + x*42;
  PRINT x;
END
```

- The corresponding AST is on the next page, generated, as usual, like this:

```
> java Sem expr.tny
> dot -Tpng sem_ast.gv > x.png
```



- As you've seen so far, all AST nodes have an attribute "env" that contains all the symbols available to that particular node.
- "env" is an **inherited attribute**, it moves down the tree.
- You can see this from the function `assignStat` that handles assignments in `Sem.java`, for example:

```
void assignStat(ASSIGN n) {  
    n.left.env = n.right.env = n.env;  
  
    expr(n.left);  
    expr(n.right);  
  
    ...  
}
```


- All the functions in Sem.java that handle expressions return the type if a subexpression in a **synthesized** attribute "type":

```
void expr(EXPR n) {  
    ...  
}  
  
void ident(IDENT n) {  
    ...  
    n.type = ...;  
}  
  
void index(INDEX n) {  
    ...  
    n.type = ...;  
}
```

```
void intlit(INTLIT n) {  
    ...  
    n.type = ...;  
}  
  
void fltlit(FLTLIT n) {  
    ...  
    n.type = ...;  
}  
  
void binop(BINOP n) {  
    ...  
    n.type = ...;  
}
```

OPTIMIZING TREEWALK EVALUATORS

- These are all the functions in Sem.java that handle expressions.
- In the following 2 tasks I want you to modify these functions so that the program makes use of fewer attributes in the AST.
- You will also need to modify the functions that call these functions!

```
void expr(EXPR n) {  
    ...  
}  
  
void ident(IDENT n) {  
    ...  
}  
  
void index(INDEX n) {  
    ...  
}  
  
void intlit(INTLIT n) {  
    ...  
}  
  
void fltlit(FLTLIT n) {  
    ...  
}  
  
void binop(BINOP n) {  
    ...  
}
```

Task 1

- Optimize the code in `Sem.java` by passing the inherited attribute `"env"` as a **function argument** instead to all the functions that handle expressions!
- Your semantic analyzer should work just the same as before, except it should not make use of any explicit assignments to the `"env"` attribute.
- Test thoroughly!
- Also, view the AST to make sure that the `env` attributes don't actually have any values assigned to them!

```
> java Sem expr.tny  
> dot -Tpng sem_ast.gv > x.png
```

Task 2

- Optimize your code by returning the "type" attribute as a **function return value**!
- Your semantic analyzer should work just the same as before, except it should not make use of any explicit assignments to the "env" attribute.
- Test thoroughly!
- Also, view the AST to make sure that the env attributes don't actually have any values assigned to them!

```
> java Sem expr.tny
```

```
> dot -Tpng sem_ast.gv > x.png
```




DEFINITIONS AND ALGORITHMS

COLLBERG.CS.ARIZONA.EDU

LIGERLABS.ORG

SUPPORTED BY
NSF SATC/TTP-1525820
SATC/EDU-2029632

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